Data Science and Management 1 (2021) 32–37

Contents lists available at ScienceDirect

Data Science and Management

journal homepage: www.keaipublishing.com/en/journals/data-science-and-management

Research article

Data science: connotation, methods, technologies, and development

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ARTICLE INFO

Keywords:
data science
data science methodology
big data
technical directions

ABSTRACT

The rapid development of big data breeds data science. Understanding and mastering the internal pattern of the value generation of big data is important for improving digitization and the covergence of data science with management science, computer science, and other disciplines. In this study, we discuss the significance of data science for the development of science and technology and social progress. Based on the interpretation of the connotation of data science, we introduce the definition of data science and review its formation by summarizing the major progress of related disciplines. We also discuss the research methodologies, development patterns and trends, and technical directions of data science. Finally, we make suggestions for data science to promote the development of data-based science and technology.

1. Introduction

The emergence and development of big data is the most direct, important, and core driving force to promote the formation of data science. The word "big" here not only means the data set contains a large amount of data, but also reflects the substantial transformation from the simple accumulation of quantity to the change of quality. Thus, the amount of data is extremely large and comprehensive that to "read the story behind the data only from these fragmented data" is possible. In addition to the massive data set, "complexity" usually refers to complex features, such as data heterogeneity, time variability, distribution, relevance, and value sparsity.

The importance of big data is that they provide the foundation for digital economy (including digital real economy, virtual economy, and network economy), scientific discovery based on data, and social governance. Big data can bring us unprecedented opportunities in at least four aspects, including management innovation, industrial development, scientific discovery, and discipline development. However, realizing the great value of big data, especially to transform big data into real productivity is still a great challenge. The challenge is mainly reflected in scientific foundation, core technology, legal system, and talent demand.

For big data, data science indicates new principles, theories, technologies, and methods. It is a new approach and problem solving method to realize the value of big data. Data science lays the foundation of big data science. It forms the core technology of big data analysis and processing, and contains an effective approach to realize the value of big data. Human society, physical space and information space are considered to be the ternary world of today's society. The relationship and interaction between the three worlds determine the technical characteristics of social development (Fig. 1).

As a big data method, the massive processing capabilities of data science (especially distributed processing capability, stream processing capability, parallel computing capability, and edge computing capability) complete the process of big data change to qualitative change. Its fusion analysis and processing capabilities (especially the interoperating technology based on the combination of virtual integration and blockchain, the heterogeneous data integration and conversion technology based on optimal transmission, etc.) realize the association and aggregation of big data; big data analysis becomes possible given the theoretically provable correctness (TPC), interpretable, generalizable, parallel, and scalable analysis algorithms. These findings indicate that data science can support the realization of big data principles, thereby empowering big data and transforming it into actual productivity and generating great value.

Peer review under responsibility of Xi'an Jiaotong University.
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https://doi.org/10.1016/j.dsm.2021.02.002
Received 7 February 2021; Accepted 8 February 2021
Available online 23 February 2021
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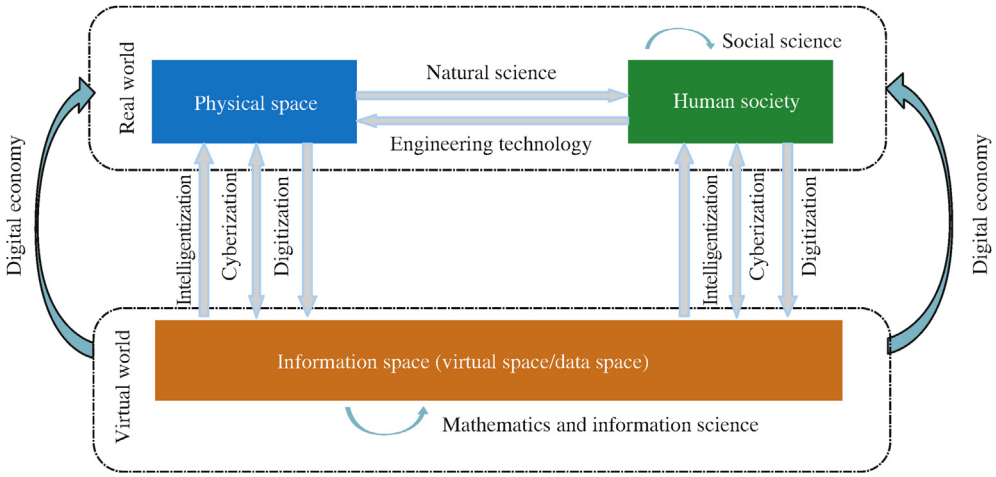


Fig. 1. Three-world theory and science technology.

2. Data science theories and methodologies

2.1. Emergence and basic concepts of data science

The emergence of data science has its inevitability, and its development has extreme urgency and importance. Data science is based on data, especially big data. The problems concerned, the methods used, and the results achieved have appeared in various forms or terms in a wide range of disciplines, such as mathematics, statistics, computer science, and artificial intelligence (AI). Data science uses the theories and methods of these related disciplines for integration, innovation, and development on this basis.

Data/Datum is a digital record of the physical world and human social activities and an information carrier in the form of encoding. Common data forms include tables, curves, graphics, images, videos, texts, audios, networks, maps, and biomics sequences. Information are data with semantic interpretation; they are valuable for decision-making after a certain abstraction and processing of the data. Information are different from matter and energy. They are "used to eliminate uncertainty".

Knowledge is obtained through the synthesis, correlation, deduction, and reasoning of information and has been verified as accurate and generally believed by people. Knowledge, often described by language and words, usually has the features of systematization and structuralization and is often described in language and text. In the field of knowledge engineering, a famous data–information–knowledge–wisdom (DIKW) hierarchy diagram depicts the hierarchical relationship from data to information, to knowledge, and then to wisdom (Rowley, 2007).

Data form a complete data chain from collection, aggregation, transmission, storage, processing, and analysis to application as shown in Fig. 2. This process is accompanied by a data value-added process from data to information, from information to knowledge, and from knowledge to decision-making. The main links of the data value chain include data collection/convergence, data storage/governance, data processing/computing, data analysis, and data application. Any data-driven science and technology realizes a specific type of data value chain.

2.2. Definition of data science

A sociological explanation of data science is science about the era of big data; it aims to reveal the new challenges, opportunities, thinking, and modes in the era of big data. It is a set of knowledge system composed of new theories, methods, models, technologies, platforms, tools, applications, and best practices in the era of big data (Li and Cheng, 2012). This explanation has sociological significance, but it is complex and lacks a substantive description of what data science should include. This discipline should be defined at least from the three dimensions of research object, methodology, and scientific task/scientific goal. On this basis, the definition of data science is introduced as follows.

Data science is the basic theory and methodology related to the realization of the data value chain. It uses modeling, analyzing, computating, and learning to study the conversion from data to information, from information to knowledge, and from knowledge to decision-making and realizes the cognition and manipulation of the real world. In data science, "realizing the value chain of data" is the research object, which is

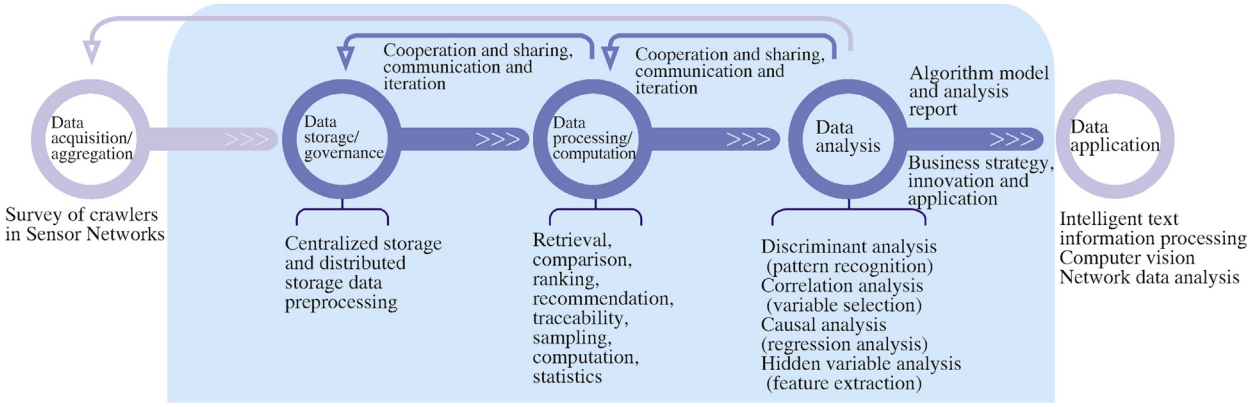


Fig. 2. Data value chain.

more accurate and targeted than "data as the research object". It can also highlight the subjectivity of big data as the research object. The goal of data science is to realize the cognition and control of the real world, and the scientific task of data science is to complete the transformation from data to information, from information to knowledge, and from knowledge to decision-making in data space, which is summarized as "three transformations and one realization". It not only limits its main connotation, but also strengthens its strong dependence on background-related disciplines. The subject methodology of data science is summarized as "a hybrid of modeling, analyzing, computating, and learning"; it is a methodological innovation. It not only affirms the multidisciplinary relevance of data science, but also avoids the simple definition of data science as a general term of multiple disciplines.

2.3. Development of the concepts of data science

Data science sprouted in the 1960s and developed from the 1990s to the beginning of this century. However, as a discipline, it is formed in the current big data era.

The embryonic stage (1960–1990). The term "datalogy" appeared in 1960–1990. In 1966, Naur, a Danish computer scientist and winner of Turing Award, proposed to define "the science of data processing" with the term "datalogy", whose research object is digital data (Naur, 1975). The emergence of the concept of datalogy provides data processing a scientific status and is the origin of its concept.

Multiperspective formation stage (1990–2010). Naur first put forward the concept of data science, and it has not been discussed for a long period. However, different subjects use different terms one after another and provide the interpretation of their respective disciplines.

New discipline establishment stage (2010–). Since the beginning of this century, big data has promoted the rapid development of data science, making it a widely accepted emerging discipline.

2.4. Relationship of data science and other disciplines

Mathematics and data science are evidently different. The former only studies abstract data (in the form of vector, matrix, and graph), but the latter focuses on entity data (such as text, image, and video). In any case, the research of data science is based on metascience, such as mathematics. Statistics and data science are almost identical in research object, research goal, and application paradigm. Statistics is not only a model of the success of data science, but also the foundation of its development. However, statistics is not a complete data science, and the differences between them are significant. First, statistics does not focus on complex data types (such as images, texts, videos, and other unstructured data), and their related applications focus on data science. Second, the research of statistics is still based on model assumptions, which still belongs to the third paradigm. Third, statistics pays less attention to the deep integration of computing science and rarely produces extensive, practical, and efficient problem-solving algorithms, such as deep learning in machine learning and AI. Fourth and most importantly, statistics are also helpless when faced with the challenges of data science, such as complex data analysis, true-false judgment, and big data analysis and processing.

Computer science is an important foundation and tool of data science. It is a theoretical foundation of data science. AI is about theory, method, and technology of simulating biological intelligence to solve problems. Data is the basis of realizing intelligence, and the mode of realizing intelligence is just the mode of data science.

AI, in a narrow sense, is one of the main contents of data science. For data science, related disciplines that portray the real world are the foundation, which is not only the basis for understanding data, the source of research questions, but also the destination of data science research. Domain-related disciplines should use data science methods, and data science should focus on discovering patterns and finding commonalities from domain-related disciplines to realize the transformation from methodology to ontology gradually.

Considering the development history of computer science, data and computing, as the two main subjects of computer processing, have been promoting each other and co-coupling development. In addition, the rapid development of computer technology (especially the Internet, cloud computing, and the Internet of Things) has greatly promoted the realization of big data value chain. The computer gradually changes from computing-centered to a new architecture of "computing + data + network", which promotes the formation and development of data science. The development is from traditional statistics to computational statistics, and exploratory data analysis, from bioinformatics to high-dimensional statistical research, from data to distributed statistical inference.

In computer science, "data" and "computation" are important research objects and the theoretical bases. In particular, with the development of big data technology, the theory, technology, and application with "data" as the core are much more focused. Its basic pursuit is the following: the "computing" technology is used to process and manage the "data" to form systematic and structured data resources, and then establish knowledge through data analysis to achieve the purpose of assisting decision-making.

Statistics is a subject that studies how to collect, organize, and analyze data effectively. It considers data as the research object and analyzes and processes data in statistical description, statistical modeling, and statistical inference. It is the most important theoretical basis and methodology of data science.

Machine learning is developed from the pattern recognition and computational learning theory in AI (Mohri et al., 2018). It mainly refers to the research and construction of a type of algorithm that enables it to learn from the data and predict the attributes without the data. Thus, machine learning overcomes the traditional and strict static program instructions, which enable the program to make decisions by using the input samples. With the focus of AI research changing from "reasoning" to "knowledge," and then to "learning," machine learning has become the most important approach to realize AI.

There has a core problem not been solved well in traditional machine learning, that is, the features of data/problems must be preset manually, in other words, the features of problems cannot be automatically extracted. Deep learning is developed to solve this problem. Deep learning is essentially a feature learning method. With deep learning, original data are transformed from some shallow model representations into deeper, more complex, and more abstract neural network representations to achieve automatic feature extraction and complete corresponding machine learning tasks. The network structure design of deep learning has always been the core of AI applications. It determines the difficulty and success rate of deep network training and the success of the application.

One of the main goals of AI is to create fully autonomous agent to optimize their own behavior and improve their ability through automatic interactive learning with the surrounding environment. Reinforcement learning is a machine learning paradigm developed to achieve this goal. The aim of federated learning is to build and train a machine learning model in collaboration with multiple computing nodes in a distributed environment. Each computing node has its own different data, and it is not allowed to exchange sample data during the training process.

2.5. Data science methodology

In data science, the description of data needs modeling, such as the random variables or probability distributions in probability theory. For data science, modeling is the formalization of objective objects and processing methods. Analyzing is a process of judging the theoretical properties of the feasibility, accuracy, complexity, and efficiency of completing the "three transformations" operation in the data space. For data science, analyzing is reasoning by using meta-knowledge. Collecting/gathering data, storing/managing data, searching/querying/

sorting/recommending data, mining/analyzing data, and making scientific discoveries based on data are performed by computers. Computer is the basic tool of data science.

For data science, algorithms are the core. Mathematics/statistics solves the problems of data science foundation and modeling, computer can achieve "accurate and fast computation", and AI helps data science to solve application problems to achieve the goal of "one realization".

For data science, learning is a tool for data empowerment. The main method of data science is not a simple synthesis or integration of modeling, analyzing, computation, and learning, but a new tool and method of data processing and analysis with theoretical accuracy and application effectiveness formed by these methods after mutual reference, interaction, infiltration, and integration. This new method integrates their own ways of thinking, expression systems, value pursuits, and core methods and has the feature of synergy. Synergy is the creation of a whole that is greater than the simple sum of the parts. It represents the "physical action" between the elements, and the effect after the "chemical action" between them, that is, the overall performance improvement after the mutual enhancement of each other. This "synergic" data science methodology has and shows great power and opens up new disciplines.

2.6. Development pattern and trend of data science

In general, data science is "growing in multiple disciplines (development), becoming famous in big data (prestige), and forming in a strong foundation (soul)". Statistics is the fastest subject to accept and strive for data science. Computer science is the first discipline to define data science from a broader perspective with great efforts. Although AI is based on big data, it is unrelated to the essential connection with data science and the possible positive reaction of data science.

Only when the following questions are understood in a scientific approach, can they be regarded as a "soul" for data science. (1) Will the data paradigm be more effective than the knowledge paradigm? (2) What is the difference between the fourth paradigm and other paradigms? (3) Better algorithms or more data? (4) Which is more important, causal analysis or correlation analysis? (5) Can query replace inference in big data application? (6) Can big data be regarded as a sample population? In summary, a rigorous data science, not a specious data subject, is required.

Data science has made extraordinary achievements in various fields. However, fundamentally, these achievements attribute more to breakthroughs in related disciplines, such as statistics, computer science, and machine learning. The essential dependence and relevance of data science is still insufficient. Data scientists hope that with its unique perspective and method, data science can develop and form more significant new scientific theories, new cognitive methods, and new application technology to exert its unique value in promoting scientific and technological progress and solving major real world problems.

Some important basic problems of science and technology in data science need to be solved, that is, exploring the data space structure and features. One of the primary tasks of data science is how to store and process heterogeneous big data, such as naturally occurring images, videos, texts, and web pages. When a set of objects is given a certain mathematical structure (operation + topology), it can become a space in a mathematical sense. The internal elements of a space in a mathematical sense can be operated according to specific rules and can also be analyzed with some specific tools. Therefore, for a type of object (such as image), standardized and strict analysis can be obtained only by considering it in the corresponding mathematical space, thereby acquiring essential recognition.

As a discipline that uses special forms of data, such as "number" and "shape" as the research object, mathematics has deeply and extensively studied all types of mathematical space, and established mathematical language, mathematical principles, and mathematical methods, which can lay the foundation of all sciences. The mature set of mathematical theory and method will be extended to data not limited to the form of numbers and shapes.

Is it possible to change "post-test" into "prediction" in the statistical process? When the size of the sample is close to the population, is sample analysis still necessary? If necessary, then what is the difference? If unnecessary, then how to perform statistical analysis directly? Is causal analysis really unnecessary? Can correlation analysis replace causal analysis? The three basic issues have been discussed at the epistemological level in recent years, but research at scientific level is rare.

Complexity and uncertainty are the most significant characteristics of big data. How to measure, describe, and use complexity and uncertainty is a basic problem of data science, and new tools and methodologies are urgently needed. It is scientific issues that must be solved to use as few samples as possible to maintain the overall characteristics more accurately, delete unimportant data safely, deal with the transformation from structured to unstructured, and sampling in real time in a data stream environment. Big data resampling should follow the "purpose-driven" principle, that is, resampling adapted to different purposes can pursue different "maintenance" goals.

3. Data science technologies

3.1. Innovation of storage computing technology and consolidation of the foundation of AI

The characteristics and effects of big data change from quantitative dimension to qualitative dimension. Thus, big data storage and computing face a revolution in basic theory and technical system. In recent years, faced with challenges, the computer science community has made great efforts in the basic theories and technological innovations of computational science and made some important progress. However, some major basic scientific problems have not yet been resolved. As a result, these basic problems restrict the development of big data storage, processing, and computing technology. In general, under the pressure of the rapid development and application of big data, computational theory, hardware architecture, system software, and application models in computer science face serious challenges.

In the big data scenario, the scale of the data itself tends to grow exponentially, resulting in the growth of computing power being less than that of data. Moreover, even for some static data, when their scale is large to a certain extent, the simple data transfer in/out has far overwhelmed the cost of computational processing. In this case, even the traditional polynomial solvable problem (P problem) cannot easily obtain the actual solution in the case of limited computing power and limited storage because a solvable problem in the small data environment may not be solvable in the big data environment. At this time, the computational complexity theory that does not consider the pattern of data scale growth in traditional computational theories must be seriously considered. From the perspective of computational theory, the dependency between data scale and model performance (upper and lower bounds) and the dependency between model complexity and model performance (upper and lower bounds) are provided. These concepts lay an important foundation for subsequent research and applications. Under these new computing paradigms, the computability theory and computational complexity theory of big data need to be reconsidered.

The integration of storage and computation is an effective idea to break through the traditional memory wall problem. Storage and computation are two completely different operators. Is there a reasonable joint optimization model? Where are the scale options and boundaries of the integration of storage and computation? How to connect the physical characteristics and process of the device seamlessly? What changes do the integration of storage and computation bring to the programming model? A breakthrough in the basic theory and architecture of the new computer architecture is necessary. How can an optimal collaboration of cloud computation-edge computing-terminal be achieved given a big data computing problem? How are complex computing tasks decomposed, and how are heterogeneous meters optimally scheduled? How is the problem of inconsistent data partition and model abstraction solved?

Big data system software especially needs to be reconstructed on the basis of basic theory and standard architecture, including theory and architecture reconstruction of big data system software, the distribution of big data analysis software, and the weakening of CAP constraints of distributed data management.

Model learning relies on a large number of training samples. How is the generalization ability of the model ensured in the case of small samples? How can the results of the model be more robust to resist attacks and deception? How are the credibility and interpretability of model inference results answered? How is a distributed and globally effective model built in a large-scale environment? How are local models used to support globally trusted model inference? How are the efficiency and accuracy of model inference balanced? These problems need to be further explored.

AI is an outstanding technology that realizes the data value chain and demonstrates the value of data. The following five basic scientific problems of AI need to be solved: (1) statistical basis of big data analysis, (2) basic algorithm of big data computation, (3) mathematical principles of deep learning, (4) optimal transport problem under unconventional constraints, and (5) study of learning methodology and learning theory in function space. The learning theory of functional space is a field that has not yet been explored. In addition to the major basic scientific problems described in the previous section, data science faces a large number of methods and technical problems to be solved. The exploration of these issues has formed some important research fields.

3.2. Technical directions of data science

Internet of Things technology. Future research on the Internet of Things needs to focus on making breakthroughs in the following areas: (1) standard issue, (2) security issue, (3) perception technology, (4) storage scheduling, and (5) complex network behavior.

Big data interoperating technology. Big data interoperating technology is necessary to realize the big data value chain. Future research needs to focus on the following aspects: (1) big data interoperating standard protocol, (2) interoperating technology under big data of low-quality, high-frequency, and high-noise, (3) software-defined big data exchange and collaborative interoperability technology, (4) data interoperating technology for ownership retention and labeling.

Big data security technology. The traditional data security protection technology cannot be entirely applied to the big data situation. Protecting big data security can start from several basic characteristics of data security, including confidentiality, integrity, usability, and controllability. In the future, the research on the following technology should be strengthened: (1) efficient data encryption for business adaptation, (2) multilevel access control of fine grain, (3) high-performance privacy protection, (4) trusted and controllable computing of distributed remote data operation, (5) confidentiality guarantee of multisource aggregation computing, and (6) big data security evaluation.

Big data storage technology. The future research of big data storage technology also needs to focus on the following aspects: (1) optimization of machine learning technology guided storage system (AI for storage), (2) domain specific big data compression and compact computing, (3) erasure coding storage native big data ecosystem, (4) big data storage method with low tail delay.

Distributed collaborative computing technology. With the development and application of technology, such as 5G communications, the Internet of Things, and big data intelligence, a large number of future computation is not limited to large-scale cloud computing centers, but are deployed to an integrated distributed resources and environment consisting of "cloud computation-edge computing-terminal". At present, the following aspects should be considered: (1) problem of adaptation autonomy in heterogeneous computing resources, (2) decision delay problem of collaboration of cloud computation-edge computing-terminal, (3) consistency problem of heterogeneous data distribution.

New database technology and system. Facing the 4V features of big data, the traditional relational database system oriented to structured data expression is incompetent for big data organization and management. For specific big data application scenarios, new database technologies designed in combination with data types and data distribution characteristics also emerge, such as in-memory databases, streaming databases, graph databases, spatio-temporal databases, and crowdsourced databases. In the future, new database theory and technology should focus on the following: (1) development of a new paradigm of database theory, (2) promotion of the development of autonomous database, (3) unified storage and management of multiple heterogeneous data, and (4) learning-based database organization model.

Big data basic algorithm. On the basis of the application background of big data, grasping the "seven giant problems" accurately is the key to big data basic algorithm research. These questions include basic statistics, generalized N-body problem, graph-theoretic computation, linear algebraic computation, optimization, integration, and alignment problem. The study of basic big data algorithms is essentially limited by the theory of big data computation. The researchers expect to find algorithms to solve problems at the level of ultralow complexity (such as at least linear complexity and below). However, when we were prepared to give up the traditional idea that the algorithm with complex polynomials is good, we suddenly begin to consider the future development of the basic algorithm of big data.

Data intelligence technology. Data intelligence technology is one of the most closely integrated fields of data science and application discipline. Its technical level represents the realization level of data value chain and is the core competence index to measure big data application. At present, almost all fields struggle to combine with deep learning, managing to promote intelligent progress in their fields. Some important areas with particular potential include the following intelligent medical care, intelligent health, intelligent education, intelligent city, and intelligent manufacturing. Some hotspots of data intelligence research also include research on machine learning theories and methods under insufficient training data, learning theories and methods that interact with the environment, sparse learning theories and methods in high-dimensional feature space, explainable and credible machine learning theories and methods, machine learning theories and methods with privacy protection, new distributed machine learning platforms for large-scale and multisource data and heterogeneous computing resources, and visual analysis technology of machine learning platforms.

Blockchain technology. The core values of blockchain are as follows. (1) It provides a trusted platform for transactions in a weak trust environment, thereby satisfying the transaction environment that people seek to reduce the cost of mutual trust. (2) It promotes the transformation of the existing Internet from information interconnection to value interconnection to form a value Internet. Blockchain is expected to realize the registration, circulation, and transaction of real world physical assets, thereby establishing a virtual society with real meaning and parallel human society. (3) It provides a credible and feasible operating platform for cognition and manipulation of the real society in parallel virtual society. Blockchain technology has been successfully applied to digital currencies and asset management, such as Bitcoin and Token. A large number of applications are found in the decentralized Internet and social governance. Blockchain is still a developing technology, and its overall development is in the pioneering period, showing an evident trend of mutual driving of technology and industrial innovation. In terms of basic theory, technology implementation, and application, blockchain still faces a series of challenges.

Big data visualization and interaction analysis technology. Generally, big data visualization and interaction technology can reveal data patterns, display data status, and adjust data processing more intuitively and directly. This approach is an important technical direction in the field of big data analysis. However, visualization and interaction analysis is similar to experimental science, which has many basic defects in formal description, theoretical model, and objective evaluation. The following principles determine the main research direction of data science: the first

is the principle of importance, that is, this direction can represent the most critical technical field for data value in the data value chain; the second is the principle of nonconflict, which indicates that the fields classified as secondary or above disciplines in other disciplines (except statistics and AI) or those generally recognized as the main direction should not be specifically listed as the direction of data science; the third is the principle of specialization, where a certain scale of practitioners is required to carry out stable and professional research to promote the sustainable development of the discipline; the fourth is the principle of integrity, where all directions can cover the data value chain and support the construction of data science knowledge system. Data collection and management mainly include data aggregation, data management, data processing, data governance, and other subdirections, which are the main part of the intersection of data science, management science, and computer science.

Data storage and computing mainly include data storage and the use of new computing tools to complete data processing tasks in the field of science and technology. Data analysis and interpretation can include three subdirections, including big data statistics, big data machine learning, and big data algorithms, which are the main manifestations of statistics and machine learning extensions. Data products and applications can mainly include big data intelligence, data product development technology and tools, and data product testing and packaging.

4. Suggestions for the development of data science

The disciplinary foundation should be consolidated to understand the value and importance of data science. The object of data science research should be grasped accurately to understand the changes in the research object. We should grasp the connotation of the basic tasks and basic scientific goals of data science. We also need to understand the essential connection between data science and digital economy. The research on major basic issues of data science should be promoted to make breakthroughs; the application includes the origin and destination.

The "data science and big data technology" should be developed to build an independent data science discipline. On the basis of the orientation of the university, the market demand, and the specialty of teachers, the appropriate subject orientation should be selected. The principle of "giving priority to charactersitics and highlighting the advantages" is the basis to determine the subject direction. We should avoid the "assorted-cold-dish" curriculum and balance the cultivation between "generalists" and "professionals". A multidisciplinary teacher team with substantial cross discipline should be established, a "hybrid" talent training mode should be adopted, and quality should be prioritized.

We should adhere to the "application demand driven and scientific problem oriented" principle and enhance the connotation and achievement abundance of data science. This approach can provide practical strategies for the implementation of national big data strategy. The focus should be big data applications in key industries/fields to propose and promote data science system solutions. Major national research tasks should be formed to organize research and make breakthroughs; in the scientific community, we should strengthen the discussion on the scientific connotation and methodology of data science, big data and AI to form a unified understanding. Among different disciplines, a consensus should be established, and the relevant concepts, theories, and methods of data science should be unified as the basis for the development of data science in different disciplines. The public's understanding of the core role of data science in the digital economy should be improved.

In addition, we shall reform the evaluation system, create an environment for interdisciplinary integration, and build a good data science development ecology. We should create an open, cooperative, inclusive, and innovative working environment. We should not enlarge the role of individual value and individual subject but advocate the value that "the inevitable factor of success is not in oneself". The cooperation culture of "mutual appreciation, integration, and cooperation" should be advocated. The performance standards of "achievement, contribution, and quality", not only "academy, ranking, or quantity" should be adopted. The distribution principle of "based on performance, considering coordination, continuity, and stability" can be used.

We shall create and cultivate a large number of data science professionals who "understand data, and can analyze and implement the measures". Accurate understanding of connotation, research direction, and methodology of data science is required. Emphasis can be the data paradigm, while adhering to the thinking mode of combining with the knowledge paradigm. We should emphasize the active attribute of data, but strengthen the normative management of data. We should be alert to "specious" views and not follow and spread them blindly. For example, we should focus on the correlation without ignoring the causality. We should also focus on the judgment of the authenticity of data and data analysis results. We should adhere to the value choice of the unity of theoretical accuracy and application efficiency and strive to improve theoretical literacy, practical ability, and scientific spirit. We should track new trends in data science, acquire new knowledge, and master new technologies, locate data science talent training goals accurately.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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